

Analyzing Mental Health: A Visualization & Machine Learning Study of DASS-21

1. Project Overview

In this project, I performed a deep dive into a dataset of ~40,000 psychometric survey results from Openpsychometrics.org. The goal was to move beyond simple data cleaning and explore how demographics and "Big Five" personality traits correlate with mental health indicators, eventually building a machine learning model to predict depressive symptoms.

2. The Data Pipeline & Integrity

I didn't take the raw data at face value. To ensure the results were reliable, I implemented several preprocessing steps:

- **The "Validity Check":** The survey included three fake words (VCL6, VCL9, VCL12). I dropped all respondents who claimed to know these words, as it suggested they weren't reading the questions.
- **Outlier Management:** I filtered out impossible age entries (e.g., people claiming to be 1,998 years old).
- **Final Sample:** After cleaning, I was left with a robust dataset of **34,576 rows**.

3. Demographics: Who is taking these tests?

The data told a very specific story about its respondents:

- **The Youth Surge:** Over 33% of respondents were aged 16–20.
- **The Myanmar Connection:** In a surprising find, **57% of the data comes from Myanmar**, which explains why 58% of the total respondents identified as Muslim and 61% as Asian.
- **Gender:** The survey skewed heavily female (77%).

4. Personality Profiling (The Big Five)

I calculated scores for the "Big Five" personality traits (Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism) using the TIPI scale.

Are Personality and Demographics Linked?

I used **Chi-Squared Tests of Independence** to see if factors like religion or education "dictated" personality traits. My findings were significant:

- **The Verdict:** I rejected all null hypotheses. Religion, sexual orientation, and education level are statistically dependent on personality traits.

- **Specific Insight:** Education level showed an incredibly strong dependency with "Openness" ($p < 0.001$), confirming that academic pursuit and openness to experience are closely linked in this cohort.

5. Predicting Depression (Machine Learning)

I set out to predict the response to **Q13: "I felt sad and depressed"** on a scale of 1 (None) to 4 (Most of the time).

Identifying the "Red Flags"

Before modeling, I ran a correlation matrix to see which survey items were the strongest predictors. The top features identified were:

- **Q26:** "I felt down-hearted and blue" (The strongest correlate).
- **Q34:** "I felt I was pretty worthless."
- **Q21:** "I felt that life wasn't worthwhile."
- **Q38:** "I felt that life was meaningless."
- **Q17:** "I felt I wasn't worth much as a person."

Model Evolution

- **Baseline (Decision Tree):** I started with a simple Decision Tree, which yielded a **55% accuracy** (a "Poor/D" grade).
- **Random Forest (Tuned):** By using validation curves to find the optimal `max_depth` (which plateaued at 9), I boosted accuracy to **67%**.

The Breakthrough: Binary Classification

I realized that the model struggled to distinguish between "Degree 3" and "Degree 4" sadness. However, in a clinical sense, the most important task is distinguishing between "Low Risk" and "High Risk."

- I binarized the target (1 & 2 = Less Likely; 3 & 4 = More Likely).
- **The Result:** The model jumped to an **"A" grade accuracy**, effectively acting as a high-performance screening tool.

6. Conclusion & Future Steps

This project proved that while demographics are interesting for context, **internal psychological self-evaluations** (like feelings of worthlessness) are the true drivers of predictive accuracy.

If I were to take this further, I would implement **SMOTE** to handle the imbalanced nature of the minority classes and experiment with **XGBoost** to see if I can achieve "A-grade" accuracy on the original 4-point scale without binarizing.

Technical Toolkit: Python, Pandas, Matplotlib, Seaborn, Scikit-Learn, Yellowbrick (for Validation Curves and Classification Reports).

Appendix: Data Codebook (DASS-42)

The 4-Point Scale

All Q-items were rated based on the past week:

1. 1 = Did not apply to me at all
2. 2 = Applied to me to some degree/some of the time
3. 3 = Applied to me to a considerable degree/good part of the time
4. 4 = Applied to me very much/most of the time

Survey Questions (Q1 - Q42)

- Q1 Upset by trivial things | Q2 Dryness of mouth | Q3 No positive feelings
- Q4 Breathing difficulty | Q5 Couldn't get going | Q6 Over-reacted
- Q7 Shakiness | Q8 Difficult to relax | Q9 Relieved when anxious situations ended
- Q10 Nothing to look forward to | Q11 Upset easily | Q12 Using nervous energy
- Q13 Felt sad and depressed (Target Variable) | Q14 Impatient with delays
- Q15 Feeling of faintness | Q16 Lost interest in everything | Q17 Not worth much as a person
- Q18 Rather touchy | Q19 Perspired noticeably | Q20 Scared without reason
- Q21 Life wasn't worthwhile | Q22 Hard to wind down | Q23 Difficulty swallowing
- Q24 No enjoyment | Q25 Heart action awareness | Q26 Down-hearted and blue
- Q27 Very irritable | Q28 Close to panic | Q29 Hard to calm down
- Q30 Feared being "thrown" by trivial tasks | Q31 Unable to be enthusiastic
- Q32 Difficult to tolerate interruptions | Q33 State of nervous tension
- Q34 Pretty worthless | Q35 Intolerant of interruptions | Q36 Terrified
- Q37 Nothing to be hopeful about | Q38 Life was meaningless | Q39 Getting agitated
- Q40 Worried about panicking | Q41 Trembling | Q42 Difficult to work up initiative

Personality (TIPI) & Demographics

- TIPI: 10 items measuring the Big Five (Extraversion, Agreeableness, Conscientiousness, Emotional Stability, Openness).
- Demographics: Includes Education (1-4), Urban/Rural childhood, Gender, Religion (12 categories), and Race.